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CBE 162: Dynamics and Control of Chemical Processes - Fall 2026

Homework 5

Due date: October 13th at 11:59 pm

A soft copy must be uploaded on Gradescope. No late homework will be accepted. Please make sure your Name and Student ID Number are in the top-right hand corner of the page. Number all pages in the bottom-right hand corner. Please describe your thought process and state all assumptions made for each question. For numerical answers, place a box around your final answer and remember units.

Problem 1 (50 points)

A chemostat is a continuous bioreactor where microrganisms convert a feed substrate S to a desired product P by some biomass, denoted by X. Accordingly, the following reaction takes place in the chemostat:

$$S + X \rightarrow P + X$$

Note that biomass is both a reactant and a product in this reaction. The reaction kinetics are modeled by monod kinetics:

$$-r_S = \mu C_x = \frac{\mu_{max} C_s}{K_m + C_s} C_x$$

To describe the formation of biomass and product, we introduce yield coefficients, which capture inefficiencies in the conversion of substrate:

$$r_X = Y_{X/S} \mu C_x = Y_{X/S} \frac{\mu_{max} C_s}{K_m + C_s} C_x$$

$$r_P = Y_{P/S} \mu C_x = Y_{P/S} \frac{\mu_{max} C_s}{K_m + C_s} C_x$$

where $Y_{X/S}$ and $Y_{P/S}$ are constant. The feed to the chemostat is sterile (no bionmass) feed of water and substrate with concentration C_{S0} and flow rate F. We assume that the reactor volume V is constant.

(a) Identify potential MV(s), DV(s), and CV(s) for this process.

(b) Develop a dynamic model for this process. *Hint: Bioreactor models are typically expressed in terms of dilution rate* $D = \frac{F}{V}$.

(c) Use the model derieved in part (b) to obtain a linear description of process dynamics around a steady-state operating point.

(d) Obtain the transfer function models mapping the inlet substrate concentration C_{S0} to the concentration of X, S, and P.

(e) Plot the process response (i.e., concentration of X, S, and P) over 30 hours for a step change in C_{S0} from $3g/L$ to $2.8g/L$ by solving the nonlinear process model from part (b), and then by using the transfer function models obtained in part (d). Overlay the responses predicted by the two models for comparison. Use the following parameters: $D_{ss} = 1hr^{-1}$, $Y_{X/S} = 0.6$, $Y_{P/S} = 0.05$, $\mu_{max} = 2hr^{-1}$, $K_m = 0.5g/L$. *Hint: You'll need to find the steady state chemostat concentrations. There are two possible solutions, ensure that your solution is non-trivial.*

(f) Plot the process response over 10 hours for a step change in C_{S0} from $3g/L$ to $0.5g/L$ by solving the nonlinear process model and the transfer function models. Use the previously defined model parameters.

(g) Based on results of parts (e) and (f), explain if the transfer function models can adequately describe the process dynamics. *Bonus: Interpret the observed process response in part (f) from a physics standpoint.*

(a) MV : F flow to reactor, CV : C_p Product Concentration, DV : C_{s0} Initial Substrate Concentration

Volumetric flow rate — Volume / sec

(b) assume constant volume

reactor well-mixed

$C_{pin} = C_{pout} = C_p$

① Specific mass/mole balance Product

mole = Volume × Concentration

$$V \cdot \frac{dC_p}{dt} =$$

$$F \cdot C_{p0} - F \cdot C_p + r \cdot V$$

$C_{p0} = 0$

$$F = \frac{dV}{dt}$$

$$D = \text{Dilution} = \frac{F}{V}$$ ↗ divide with volume

$$\frac{dC_p}{dt} = -D \cdot C_p - r = -D C_p + Y_{P/S} \cdot \mu_{max} \frac{C_s}{K_m + C_s} C_x$$

② Substrate balance

$$V \cdot \frac{dC_s}{dt} = F \cdot C_{s0} - F C_s + V \cdot r_s$$

substrate consumption rate

$$\frac{dC_s}{dt} = D \cdot (C_{s0} - C_s) - \mu_{max} \frac{C_s}{K_m + C_s} \cdot C_x$$

③ Biomass Balance

$$V\frac{dC_X}{dt} = F\cancel{C_{X0}} - FC_X + V\cdot r_X$$

(XX)

$$\frac{dC_X}{dt} = D\cdot(-C_X) + Y_{X/S}\cdot\mu_{max}\frac{C_S}{k_m + C_S}C_X$$

④

$$\tilde{C_S} = C_S - C_{S,SS}, \quad \tilde{C_X} = C_X - C_{XSS}$$

Deviation Term

$$\tilde{C_P} = C_P - C_{PSS}, \quad \tilde{F} = F - F_{SS}, \quad \tilde{C_{S0}} = C_{S0} - C_{S0,SS}.$$

f: linear description $f = \frac{d\tilde{C_S}}{dt}$ $f(X_{SS}) = \sum \frac{f^n(X_{SS})}{n!}(X - X_{SS})^n$

Taylor series expansion to approach deviation term

For substrate $f = \frac{d\tilde{C_S}}{dt}$, multi-variable Taylor expansion.

$$\frac{dC_S}{dt} = f(D, C_{S0}, C_X, C_S)$$

evaluated at steady state $= \tilde{D}$

$$f \approx \cancel{f|_{SS}}^{0} + \frac{\partial f}{\partial D}\Big|_{SS}(D - D_{SS}) + \frac{\partial f}{\partial C_{S0}}\Big|_{SS}\tilde{C_{S0}} + \frac{\partial f}{\partial C_X}\tilde{C_X} + \frac{\partial f}{\partial C_S}\tilde{C_S}$$

(new value = original value + slope × change in input)

↓

$$f = \left(C_{S0,ss} - C_{S,ss}\right)\tilde{D} + D_{ss}\cdot\tilde{C}_{S0} + \frac{-\mu_{max}\cdot C_{S,ss}}{K_m + C_{S,ss}}\cdot\tilde{C}_X$$

$$-\left(D - \mu_{max}C_X\cdot\frac{d}{dC_S}\left(\frac{C_S}{K_m + C_S}\right)\right)\cdot\tilde{C}_S$$

Quotient Rule: $h(x) = \frac{f(x)}{g(x)}$

$h'(x) = \frac{f'g - fg'}{g(x)^2}$

$$-\left(D_{ss} - \mu_{max}C_X\cdot\frac{[K_m + C_S] - [C_S]}{(K_m + C_{S,ss})^2}\right)\tilde{C}_S$$

$f(x) = C_S \quad \frac{f(x)}{dx} = 1$

$g(x) = K_m + C_S \quad \frac{g(x)}{dx} = 1$

$$\frac{d\tilde{C}_S}{dt} \simeq \left(C_{S0,ss} - C_{S,ss}\right)\cdot\tilde{D} + D_{ss}\cdot\tilde{C}_{S0} - \frac{\mu_{max}C_{S,ss}}{K_m + C_{S,ss}}\tilde{C}_X$$

$$-D_{ss}\cdot\tilde{C}_S - \frac{\mu_{max}\cdot C_X\cdot\tilde{C}_S}{(K_m + C_{S,ss})^2}$$

#

$(-D_{ss} + D_{ss}) = 0$

$$\frac{d\tilde{C}_X}{dt} \simeq \cancel{\frac{d\bar{C}_X}{dt}\Big|_{ss}}^{0} - C_{X,ss}\tilde{D} - \left(D_{ss}\tilde{C}_X + Y_{X/S}\frac{\mu_{max}\,C_{S,ss}}{K_m + C_{S,ss}}\right)\cdot\tilde{C}_X$$

$$+ Y_{X/S}\frac{\mu_{max}\,C_{X,ss}}{(K_m + C_{S,ss})^2}\cdot\tilde{C}_S$$

$$\frac{d\bar{C}_X}{dt}\Big|_{ss} = 0 = -D_{ss}C_{X,ss} + Y_{X/S}\frac{\mu_{max}C_{S,ss}}{K_m + C_{S,ss}}C_{X,ss}$$

$$D_{ss} = Y_{X/S}\frac{\mu\,C_{S,ss}}{K_m + C_{S,ss}}$$

#

$$\rightarrow \frac{d\tilde{C}_X}{dt} = -C_{X,ss}\tilde{D} + Y_{X/S}\frac{\mu_{max}C_{S,ss}}{(K_m + C_{S,ss})^2}\cdot\tilde{C}_S$$

$$\frac{d\tilde{C}_P}{dt} \simeq -C_{P,ss}\tilde{D} - \tilde{C}_P\cdot D_{ss} + Y_{P/S}\frac{\mu_{max}C_{S,ss}}{K_m + C_{S,ss}}\tilde{C}_X$$

$$+ Y_{P/S}\cdot\frac{\mu_{max}C_{X,ss}}{(K_m + C_{S,ss})^2}\tilde{C}_S$$

#

d)

$$\frac{d\tilde{C}_S}{dt} = (C_{S0,ss} - C_{S,ss})\tilde{D} + D_{ss}\cdot\tilde{C}_{S0} - \frac{\mu_{max} C_{S,ss}}{K_m + C_{S,ss}}\tilde{C}_X - \left(D_{ss} + \frac{\mu_{max} K_m \cdot C_{X,ss}}{(K_m + C_{S,ss})^2}\right)\tilde{C}_S$$

← laplace

$$\rightarrow \quad s\cdot\tilde{C}_S(s) = (C_{S0} - C_{S,ss})\tilde{D}(s) + D_{ss}\tilde{C}_{S0} - \frac{\mu_{max} C_{S,ss}}{K_m + C_{S,ss}}\tilde{C}_X(s)$$

$$- D_{ss} + \frac{\mu_{max}\cdot K_m \cdot C_{X,ss}}{(K_m + C_{S,ss})^2}\tilde{C}_S(s)$$

move $\tilde{C}_S(s)$

$$\tilde{C}_S(s) = \frac{1}{s + D_{ss} + \frac{\mu_{max} K_m C_{X,ss}}{(K_m + C_{S,ss})^2}} \times \left[(C_{S0,ss} - C_{S,ss})\tilde{D}(s) + D_{ss}\tilde{C}_{S0}(s) - \frac{\mu_{max} C_{S,ss}}{K_m + C_{S,ss}}\tilde{C}_X(s)\right]$$

#

$$\tilde{C}_X = \frac{1}{s}\cdot\left(-C_{X,ss}\tilde{D} + Y_{X/S}\frac{\mu_{max} C_{X,ss}}{(K_m + C_{S,ss})^2}\tilde{C}_S\right)$$

#

Transfer function for change in C_{so}?

$C_{so}(s) \rightarrow C_s(s) \rightarrow C_x(s)$

Feed input — Substrate — Biomass

$\tilde{D}(s) = 0$, $\tilde{C}_x(s) \neq 0$

$$a = D_{ss} + \frac{\mu \cdot k \cdot C}{(k+C)^2} \quad , \quad b = \frac{\mu \cdot C}{k+C} \quad , \quad c = Y \frac{\mu \cdot k \cdot C}{(k+C)^2}$$

Transform equation

$$(s+a) \cdot \tilde{C}_s(s) + b\, \tilde{C}_x(s) = D_{ss}\, \tilde{C}_{so}(s)$$

$$s\, \tilde{C}_x(s) = c \cdot \tilde{C}_s(s)$$

$$(s^2 + as + bs)\, \tilde{C}_s(s) = D_{ss} \cdot s \cdot \tilde{C}_{so}(s)$$

↳ $G_s(s) = \frac{\tilde{C}_s(s)}{\tilde{C}_{so}(s)} = \frac{D_{ss} \cdot s}{s^2 + as + bc}$ #

Substrate C_s

$$G_X(s) = \frac{\tilde{C}_X(s)}{\tilde{C}_{S0}(s)} = \frac{C \cdot D_{ss}}{s^2 + as + bC}$$

#

Biomass C_X

$$p = Y \cdot \frac{\mu \cdot C_{S,ss}}{K + C_{S,ss}} \qquad q = Y \cdot \frac{\mu \cdot K \cdot C_{X,ss}}{(K + C_{S,ss})^2}$$

#

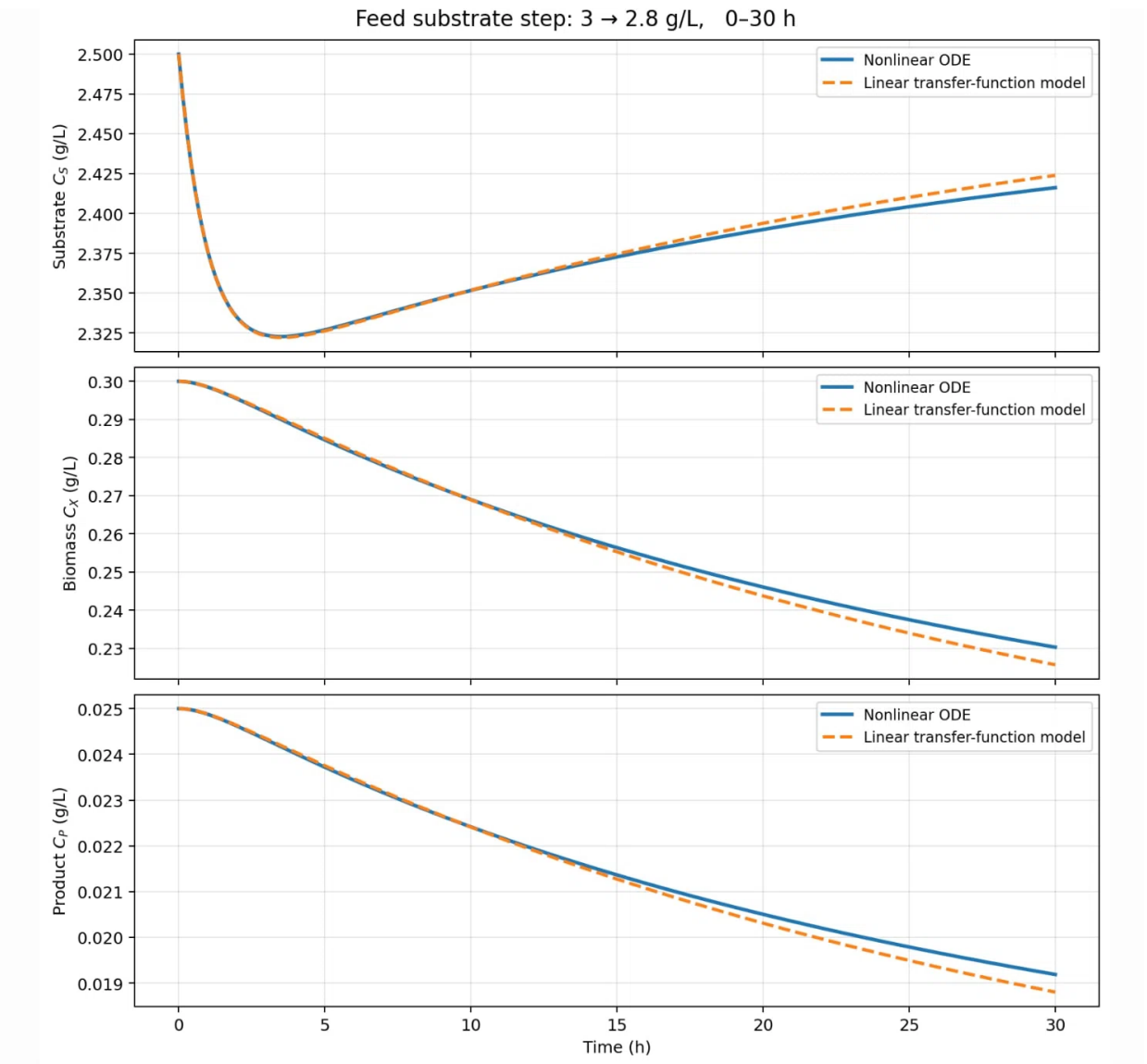
$$G_P(s) = \frac{\tilde{C}_P(s)}{C_{S0}(s)} = \frac{D_{ss}(pC + qs)}{(s + D_{ss}) \cdot (s^2 + as + bC)}$$

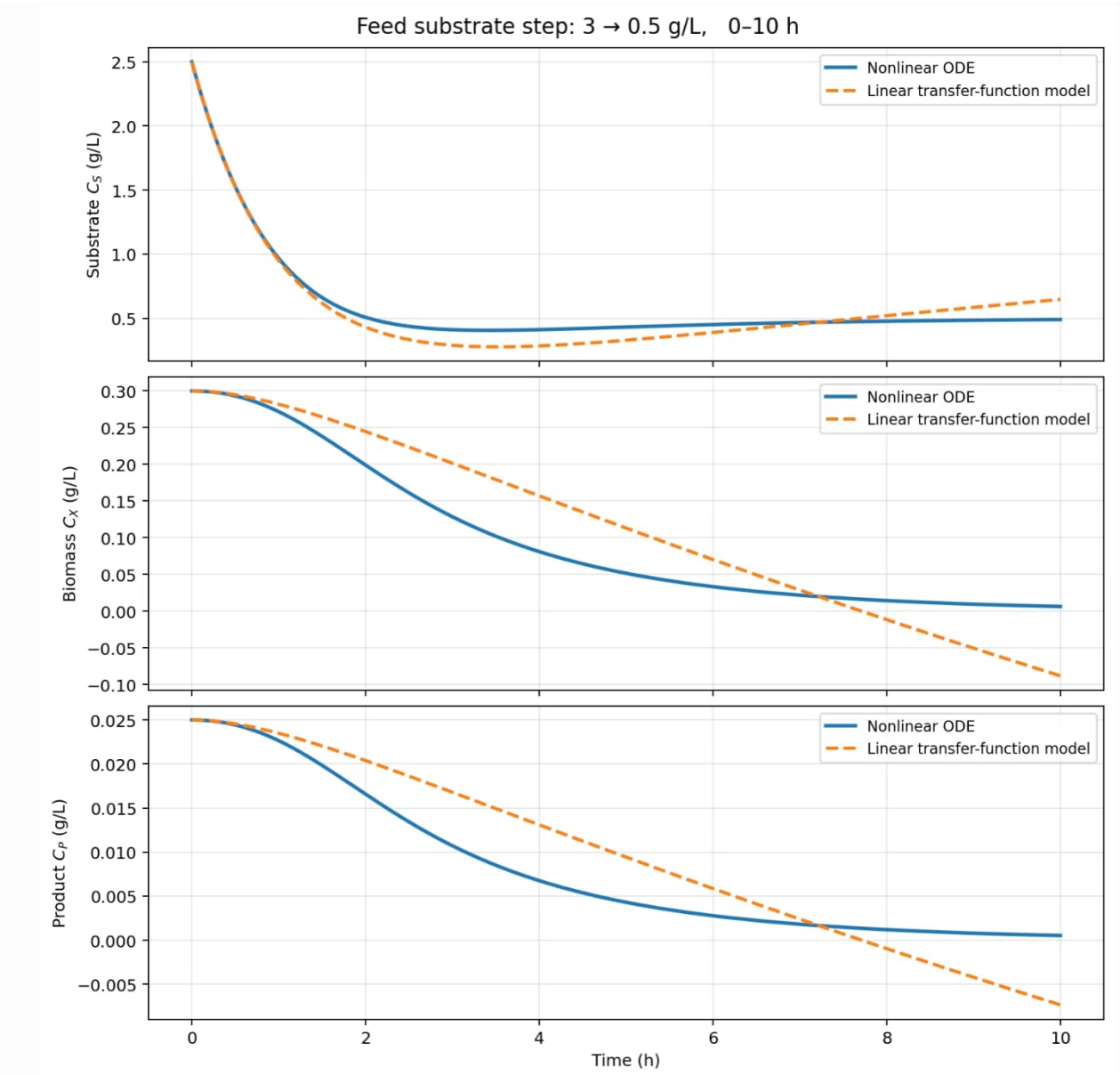
Product C_P

(e) $C_{S,ss} = 2.5$, $C_{X,ss} = 0.3$, $C_{P,ss} = 0.025$ g/L

$C_{S0} = 3 \rightarrow 2.8$ g/L, for 30 hours

(f) $C_{S0} = 3 \rightarrow 0.5$ g/L, for 10 hours




Feed substrate step: 3 → 0.5 g/L, 0–10 h
Substrate C_S (g/L)
Biomass C_X (g/L)
Product C_P (g/L)
Time (h)
Nonlinear ODE
Linear transfer-function model
2.5
2.0
1.5
1.0
0.5
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
0
2
4
6
8
10

⑨ linear model good in small perturbation process.

$\Delta C_{s0} = 3 \rightarrow 2.8\ g/L$

first order Taylor series remains valid.

Large change,

non-linear model will be better!

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physics for Chemostat:

$$\frac{dC_x}{dt} = \underbrace{Y_{x/s} \frac{\mu_{max} C_s}{K_m + C_s} C_x}_{\text{Growth rate}} - \underbrace{D C_x}_{\text{washout}}$$

⑩



Sub: $3 \rightarrow 2.8$ g/L

Inlet substrate at steady state decrease

→ growth rate decrease

- dilution rate D is constant



↓

so bio mass decrease

↓

substrate Consumption decrease

↓

not sub input

$$\frac{dC_S}{dt} = D(C_{S0} - C_S) - \frac{\mu C_S}{K + C_S} \cdot C_X$$

reactor Substrate recovers

↓

new steady sta

↓

Substrate consumption ↓

f) C_{S0} 3 → 0.5 not enough

$C_S = 2.5$ g/L to maintain biomass in steady state

$$\left(Y \frac{\mu \cdot C}{K+C} = D \rightarrow 0.6 \cdot \frac{2 \cdot C}{0.5+C} = 1\right.$$

$$C = 2 \text{ g/l}$$

→ Cell growth slower than removal by wash out

→ Biomass ↓ to 0 , Product formation ↓ to 0

→ Substrate Concentration → feed Concentration 0.5 g/l

$$C_{S,ss} = 0.5 \text{ g/l}, \quad C_{X,ss} = 0 \text{ g/l}, \quad C_{P,ss} = 0 \text{ g/l}$$